Asymptotic Normality of a Centered LDSC Estimator under Fixed Genetic Effects

Model, estimator, and target parameter

Let $X \in \mathbb{R}^{n\times p}$ be the genotype matrix in the GWAS cohort, let $y \in \mathbb{R}^n$ be the outcome vector, and let $Z \in \mathbb{R}^{n_r\times p}$ be an independent reference panel. We assume throughout that the rows

$$x_1, \dots, x_n \in \mathbb{R}^p, \qquad z_1, \dots, z_{n_r} \in \mathbb{R}^p,$$

are independent and identically distributed Gaussian vectors with mean zero and common covariance matrix $\Sigma \in \mathbb{R}^{p\times p}$. The GWAS cohort and the reference panel are independent. The outcome obeys the linear model

$$y = X\alpha + \varepsilon,$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top \in \mathbb{R}^p$ is an unknown but *fixed* vector of genetic effects, and

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^\top \sim N(0, \sigma_\varepsilon^2 I_n)$$

is independent of X and Z. We stress that no random-effects assumption is imposed on α.

For definiteness, we define the reference-panel covariance estimator by

$$\widehat{\Sigma} := \frac{1}{n_r} Z^\top Z.$$

(The same proof goes through for the centered unbiased sample covariance, with only $O_p(n_r^{-1})$ changes in lower-order terms.) For $j \in \{1, \dots, p\}$ define

$$\widehat{\ell}_j := \sum_{i=1}^{p} \widehat{\Sigma}_{ji}^2 = (\widehat{\Sigma}^2)_{jj}, \qquad \widehat{a}_j := \frac{1}{n} X_j^\top y, \qquad \widehat{w}_j := \widehat{a}_j^2,$$

where X_j denotes the jth column of X. Put

$$\widehat{\ell} = (\widehat{\ell}_1, \dots, \widehat{\ell}_p)^\top, \qquad \widehat{w} = (\widehat{w}_1, \dots, \widehat{w}_p)^\top,$$

and let

$$\mu_{\widehat{\ell}} := \frac{1}{p}\mathbf{1}^\top \widehat{\ell}, \qquad \widehat{\mu}_w := \frac{1}{p}\mathbf{1}^\top \widehat{w}, \qquad P := I_p - \frac{1}{p}\mathbf{1}\mathbf{1}^\top.$$

Then the centered LDSC slope estimator is

$$\widehat{\sigma}_\alpha^2 := \frac{(\widehat{\ell} - \mu_{\widehat{\ell}}\mathbf{1})^\top (\widehat{w} - \widehat{\mu}_w \mathbf{1})}{(\widehat{\ell} - \mu_{\widehat{\ell}}\mathbf{1})^\top (\widehat{\ell} - \mu_{\widehat{\ell}}\mathbf{1})} = \frac{(P\widehat{\ell})^\top (P\widehat{w})}{\left\| P\widehat{\ell} \right\|_2^2}.$$

We now define the deterministic population target. Let

$$\ell := \big((\Sigma^2)_{11}, \dots, (\Sigma^2)_{pp}\big)^\top, \qquad b := \Sigma\alpha, \qquad m := b^{\circ 2} = (b_1^2, \dots, b_p^2)^\top,$$

and set

$$c := P\ell, \qquad Q := \|c\|_2^2 .$$

Whenever $Q > 0$, define

$$\sigma_\alpha^2 := \frac{c^\top m}{Q} = \frac{c^\top P m}{Q}. \tag{1}$$

Thus σ_α^2 is the coefficient of the orthogonal projection of the centered vector $m - P^\perp m$ onto the centered LD-score vector c. This is the natural fixed-effect target corresponding to the centered LDSC regression.

Assumptions

We impose the following high-dimensional conditions.

Assumption 1 (Block-diagonal Gaussian design)**.** For each p, the covariance matrix $\Sigma = \Sigma_p$ is block diagonal,

$$\Sigma = \operatorname{diag}\big(\Sigma^{(1)}, \dots, \Sigma^{(B_p)}\big),$$

with block sizes bounded uniformly in p:

$$\max_{1 \le b \le B_p} \dim\big(\Sigma^{(b)}\big) \le D$$

for some finite constant D not depending on p.

Assumption 2 (Uniform spectral bounds and standardization)**.** The diagonal entries of Σ satisfy $\Sigma_{jj} = 1$ for all j, and there exist constants $0 < \lambda_- \le \lambda_+ < \infty$ such that, for every block $\Sigma^{(b)}$,

$$\lambda_- \le \lambda_{\min}(\Sigma^{(b)}) \le \lambda_{\max}(\Sigma^{(b)}) \le \lambda_+.$$

Assumption 3 (Fixed-effect bounded signal)**.** The vector $\alpha = \alpha_p \in \mathbb{R}^p$ is deterministic and satisfies

$$\alpha^\top \Sigma \alpha \le M$$

for some finite constant M independent of p.

Assumption 4 (Joint high-dimensional regime)**.** As $n, p, n_r \to \infty$,

$$\frac{p}{n} \to \gamma \in (0, \infty), \qquad \frac{p}{n_r} \to \gamma_r \in (0, \infty).$$

Assumption 5 (Nondegenerate LD-score variation)**.** There exists $q_\ell \in (0, \infty)$ such that

$$\frac{Q}{p} = \frac{1}{p}\,\|P\ell\|_2^2 \longrightarrow q_\ell.$$

To state the last assumption, define

$$d := Pm - 2\sigma_\alpha^2 c.$$

Further, define the linearized statistic

$$L_p := \frac{1}{Q}\, c^\top P\big(\widehat{w} - \mathbb{E}\widehat{w}\big) + \frac{1}{Q}\,\big(P(\widehat{\ell} - \mathbb{E}\widehat{\ell})\big)^\top d. \tag{2}$$

Assumption 6 (Nondegenerate first-order variance)**.** There exist constants $0 < v_- \le v_+ < \infty$ such that

$$v_- \le p\ \mathrm{Var}(L_p) \le v_+$$

for all sufficiently large p.

Remark 1. Assumption 6 is the only nondegeneracy hypothesis not implied directly by the design assumptions. It is the natural requirement that the first-order fluctuation of the centered LDSC slope be genuinely of order $p^{-1/2}$. In concrete block models it can be checked by explicit moment calculations, using the same Wishart identities employed below.

Main theorem

Theorem 1 (Asymptotic normality of the centered LDSC estimator)**.** *Under Assumptions 1–6, the following statements hold.*

(i) The estimator is asymptotically unbiased for the fixed-effect target (1) *up to order* p^{-1}*:*

$$\mathbb{E}\widehat{\sigma}^2_\alpha = \sigma^2_\alpha + O\left(\frac{1}{n} + \frac{1}{n_r}\right) = \sigma^2_\alpha + O\left(\frac{1}{p}\right).$$

(ii) The centered and variance-normalized estimator is asymptotically standard normal:

$$\frac{\widehat{\sigma}^2_\alpha - \mathbb{E}\widehat{\sigma}^2_\alpha}{\sqrt{\mathrm{Var}(\widehat{\sigma}^2_\alpha)}} \Longrightarrow N(0,1).$$

(iii) If, in addition, $p\,\mathrm{Var}(L_p) \to v^2 \in (0,\infty)$*, then*

$$\sqrt{p}\left(\widehat{\sigma}^2_\alpha - \sigma^2_\alpha\right) \Longrightarrow N(0, v^2).$$

Proof of Theorem 1

The proof is organized into several steps.

Step 1: deterministic target and exact mean formulas

Let

$$\tau^2 := \alpha^\top \Sigma \alpha + \sigma^2_\varepsilon.$$

Since $y_i = x_i^\top \alpha + \varepsilon_i$, each pair (X_{ij}, y_i) is jointly centered Gaussian with

$$\mathbb{E}(X_{ij} y_i) = e_j^\top \Sigma \alpha = b_j, \qquad \mathrm{Var}(X_{ij}) = 1, \qquad \mathrm{Var}(y_i) = \tau^2.$$

By Isserlis' formula,

$$\mathbb{E}(X^2_{ij} y_i^2) = \mathrm{Var}(X_{ij})\,\mathrm{Var}(y_i) + 2\,\mathrm{Cov}(X_{ij}, y_i)^2 = \tau^2 + 2b_j^2.$$

Therefore

$$\begin{aligned}\mathbb{E}(\widehat{a}_j^2) &= \mathbb{E}\left[\frac{1}{n^2}\sum_{r=1}^{n}\sum_{s=1}^{n} X_{rj}y_r X_{sj}y_s\right] \\ &= \frac{1}{n^2}\Big[n\,\mathbb{E}(X_{1j}^2y_1^2) + n(n-1)\,\mathbb{E}(X_{1j}y_1)^2\Big] \\ &= \frac{1}{n^2}\Big[n(\tau^2+2b_j^2) + n(n-1)b_j^2\Big] \\ &= \left(1+\frac{1}{n}\right)b_j^2 + \frac{\tau^2}{n}.\end{aligned}$$

Hence

$$\mathbb{E}\widehat{w} = \left(1+\frac{1}{n}\right)m + \frac{\tau^2}{n}\mathbf{1}, \qquad P\mathbb{E}\widehat{w} = \left(1+\frac{1}{n}\right)Pm. \tag{3}$$

Next we compute $\mathbb{E}\widehat{\ell}$. Let $W_r := Z^\top Z$, so that $W_r \sim \text{Wishart}(n_r, \Sigma)$ and

$$\widehat{\Sigma} = \frac{1}{n_r}W_r, \qquad \widehat{\ell}_j = \frac{1}{n_r^2}e_j^\top W_r^2 e_j.$$

For a Wishart matrix $W \sim \text{Wishart}(m, \Sigma)$ and a deterministic vector u, Proposition S1 of Dicker's supplement gives the identity

$$\mathbb{E}(u^\top W^2 u) = m\ \text{tr}(\Sigma)\,u^\top\Sigma u + m(m+1)u^\top\Sigma^2 u.$$

Applying this with $m = n_r$ and $u = e_j$, and using $\text{tr}(\Sigma) = p$ and $\Sigma_{jj} = 1$, we obtain

$$\mathbb{E}(e_j^\top W_r^2 e_j) = n_r p + n_r(n_r+1)(\Sigma^2)_{jj} = n_r p + n_r(n_r+1)\ell_j.$$

Therefore

$$\mathbb{E}\widehat{\ell}_j = \frac{p}{n_r} + \left(1+\frac{1}{n_r}\right)\ell_j, \qquad P\mathbb{E}\widehat{\ell} = \left(1+\frac{1}{n_r}\right)c. \tag{4}$$

Consequently,

$$\theta_p := \frac{(P\mathbb{E}\widehat{\ell})^\top(P\mathbb{E}\widehat{w})}{\left\|P\mathbb{E}\widehat{\ell}\right\|_2^2} = \frac{1+n^{-1}}{1+n_r^{-1}}\,\sigma_\alpha^2 = \sigma_\alpha^2 + O\left(\frac{1}{n}+\frac{1}{n_r}\right). \tag{5}$$

This already yields the correct first-order centering.

Step 2: first-order expansion of the ratio

Define centered fluctuations

$$\delta_w := P\big(\widehat{w} - \mathbb{E}\widehat{w}\big), \qquad \delta_\ell := P\big(\widehat{\ell} - \mathbb{E}\widehat{\ell}\big).$$

By (3)–(4),

$$P\widehat{w} = \left(1+\frac{1}{n}\right)Pm + \delta_w, \qquad P\widehat{\ell} = \left(1+\frac{1}{n_r}\right)c + \delta_\ell. \tag{6}$$

Let

$$\bar{c} := \left(1+\frac{1}{n_r}\right)c, \qquad \bar{m} := \left(1+\frac{1}{n}\right)Pm, \qquad \bar{Q} := \|\bar{c}\|_2^2.$$

Then

$$\widehat{\sigma}_\alpha^2 = \frac{(\bar{c}+\delta_\ell)^\top(\bar{m}+\delta_w)}{\|\bar{c}+\delta_\ell\|_2^2}.$$

Consider the map

$$F(u,v) := \frac{u}{v}, \qquad v \neq 0.$$

A Taylor expansion of F at

$$u_0 := \bar{c}^\top \bar{m}, \qquad v_0 := \bar{Q}$$

shows that

$$\widehat{\sigma}_\alpha^2 - \theta_p = \frac{1}{\bar{Q}}\,\bar{c}^\top\delta_w + \frac{1}{\bar{Q}}\,\delta_\ell^\top(\bar{m} - 2\theta_p\bar{c}) + r_p, \tag{7}$$

where

$$|r_p| \le C\left(\frac{\|\delta_\ell\|_2\,\|\delta_w\|_2}{\bar{Q}} + \frac{\|\delta_\ell\|_2^2}{\bar{Q}}\right) \tag{8}$$

for a deterministic constant C.

We now show that r_p is negligible. Because the block size is uniformly bounded and all eigenvalues of the blocks are uniformly bounded, every entry of x_i and z_i belongs to a finite-dimensional Gaussian system with moments bounded uniformly in p. Since Assumption 3 implies $\alpha^\top\Sigma\alpha \le M$, all moments of y_i are uniformly bounded as well. Hence there is a constant C such that

$$\sup_{j\le p} \mathrm{Var}(\widehat{a}_j^2) \le \frac{C}{n}.$$

Similarly, the same Wishart moment identities as in Proposition S1 imply

$$\sup_{j\le p} \mathrm{Var}(\widehat{\ell}_j) \le \frac{C}{n_r}.$$

The block-diagonal structure implies that each coordinate $\widehat{a}_j^2$ is correlated with only $O(1)$ other coordinates, and the same holds for $\widehat{\ell}_j$; therefore

$$\mathbb{E}\,\|\delta_w\|_2^2 \le C\frac{p}{n} = O(1), \qquad \mathbb{E}\,\|\delta_\ell\|_2^2 \le C\frac{p}{n_r} = O(1). \tag{9}$$

By Assumption 5,

$$\bar{Q} = \left(1+\frac{1}{n_r}\right)^2 Q \asymp p.$$

Combining this with (8) and (9), we obtain

$$r_p = O_p(p^{-1}). \tag{10}$$

Since

$$\bar{c} - c = O(p^{-1}) \quad \text{coordinatewise}, \qquad \bar{m} - 2\theta_p\bar{c} - d = O(p^{-1}) \quad \text{coordinatewise},$$

and $\|\delta_w\|_2 + \|\delta_\ell\|_2 = O_p(1)$ by (9), equation (7) yields

$$\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2 = L_p + b_p + o_p(p^{-1/2}), \tag{11}$$

where

$$b_p := \theta_p - \sigma_\alpha^2 = O\left(\frac{1}{n}+\frac{1}{n_r}\right) = O(p^{-1}). \tag{12}$$

In particular,

$$\mathbb{E}\widehat{\sigma}_\alpha^2 = \sigma_\alpha^2 + O\left(\frac{1}{n}+\frac{1}{n_r}\right),$$

which proves part (i) of the theorem.

Step 3: representation of the linear term as a smooth Gaussian functional

We next prove asymptotic normality of L_p. Write

$$C_p := \frac{1}{Q}\,\mathrm{diag}(c), \qquad D_p := \frac{1}{Q}\,\mathrm{diag}(d).$$

Then

$$\frac{1}{Q}\,c^\top \delta_w = \frac{1}{n^2} y^\top X C_p X^\top y - \mathbb{E}\left[\frac{1}{n^2} y^\top X C_p X^\top y\right], \tag{13}$$

because

$$c^\top P\widehat{w} = c^\top \widehat{w} = \sum_{j=1}^{p} c_j \widehat{a}_j^2 = \frac{1}{n^2} y^\top X\,\mathrm{diag}(c) X^\top y.$$

Likewise,

$$\frac{1}{Q}\,\delta_\ell^\top d = \frac{1}{n_r^2}\,\mathrm{tr}(D_p W_r^2) - \mathbb{E}\left[\frac{1}{n_r^2}\,\mathrm{tr}(D_p W_r^2)\right]. \tag{14}$$

Therefore

$$L_p = g_{1,p}(X,\varepsilon) + g_{2,p}(Z), \tag{15}$$

where

$$g_{1,p}(X,\varepsilon) := \frac{1}{n^2} y^\top X C_p X^\top y - \mathbb{E}\left[\frac{1}{n^2} y^\top X C_p X^\top y\right],$$

$$g_{2,p}(Z) := \frac{1}{n_r^2}\,\mathrm{tr}(D_p W_r^2) - \mathbb{E}\left[\frac{1}{n_r^2}\,\mathrm{tr}(D_p W_r^2)\right].$$

By Assumption 5,

$$\|C_p\|_{\mathrm{op}} = O(p^{-1}), \qquad \|D_p\|_{\mathrm{op}} = O(p^{-1}). \tag{16}$$

Step 4: derivative bounds for the Gaussian second-order Poincaré inequality

We invoke the Gaussian second-order Poincaré inequality in the form of Chatterjee's theorem: if $G = G(U)$ is a smooth functional of a centered Gaussian vector U, then the total variation distance between the law of G and the Gaussian law with the same variance is controlled by the size of the first and second derivatives of G. We verify the requisite derivative bounds for L_p.

The GWAS term. Let $\widetilde{U} = (u, U)$ with $u \in \mathbb{R}^n$, $U \in \mathbb{R}^{n\times p}$, and

$$\|u\|_2^2 + \|U\|_F^2 = 1.$$

Define

$$q := U\alpha + u.$$

Since $\|U\|_{\mathrm{op}} \le \|U\|_F \le 1$ and Assumption 3 together with Assumption 2 implies $\|\alpha\|_2 = O(1)$, we have

$$\|q\|_2 \le \|U\|_{\mathrm{op}}\,\|\alpha\|_2 + \|u\|_2 = O(1). \tag{17}$$

Consider the perturbation

$$X_t := X + tU, \qquad y_t := y + tq, \qquad \phi(t) := \frac{1}{n^2} y_t^\top X_t C_p X_t^\top y_t.$$

Then the first and second directional derivatives of $g_{1,p}$ in the direction $\widetilde{U}$ are $\phi'(0)$ and $\phi''(0)$ respectively. Direct differentiation gives

$$\phi'(0) = \frac{2}{n^2}\Big(y^\top U C_p X^\top y + q^\top X C_p X^\top y\Big),$$

whence

$$\big|\phi'(0)\big| \le \frac{C\,\|C_p\|_{\mathrm{op}}}{n^2}\Big(\|X\|_{\mathrm{op}}\,\|y\|_2^2 + \|X\|_{\mathrm{op}}^2\,\|y\|_2\Big). \tag{18}$$

Similarly,

$$\phi''(0) = \frac{2}{n^2}\Big(y^\top U C_p U^\top y + 2q^\top U C_p X^\top y + 2q^\top X C_p U^\top y + q^\top X C_p X^\top q\Big),$$

so that

$$\big|\phi''(0)\big| \le \frac{C\,\|C_p\|_{\mathrm{op}}}{n^2}\Big(\|y\|_2^2 + \|X\|_{\mathrm{op}}\,\|y\|_2 + \|X\|_{\mathrm{op}}^2\Big). \tag{19}$$

For a Gaussian matrix with covariance uniformly bounded in operator norm and with $p/n \to \gamma \in (0,\infty)$, standard random-matrix bounds imply that $p^{-1/2}\,\|X\|_{\mathrm{op}}$ has moments of all orders bounded uniformly in p. Likewise, because $\tau^2 = \alpha^\top \Sigma \alpha + \sigma_\varepsilon^2 = O(1)$, the quantity $n^{-1/2}\,\|y\|_2$ has moments of all orders bounded uniformly in p. Combining these facts with (16), (18), and (19), we obtain

$$\Big(\mathbb{E}\,\|\nabla g_{1,p}\|_2^4\Big)^{1/4} = O(p^{-1/2}), \qquad \Big(\mathbb{E}\,\big\|\nabla^2 g_{1,p}\big\|_{\mathrm{op}}^4\Big)^{1/4} = O(p^{-1}). \tag{20}$$

The reference-panel term. Let $U \in \mathbb{R}^{n_r \times p}$ satisfy $\|U\|_F = 1$. Define

$$Z_t := Z + tU, \qquad W_t := Z_t^\top Z_t = W_r + tA + t^2 B,$$

where

$$A := U^\top Z + Z^\top U, \qquad B := U^\top U.$$

Set

$$\psi(t) := \frac{1}{n_r^2}\,\mathrm{tr}(D_p W_t^2).$$

Then

$$\psi'(0) = \frac{2}{n_r^2}\,\mathrm{tr}(D_p W_r A),$$

so that

$$\big|\psi'(0)\big| \le \frac{C\,\|D_p\|_{\mathrm{op}}}{n_r^2}\,\|W_r\|_{\mathrm{op}}\,\|A\|_{\mathrm{op}} \le \frac{C}{n_r^2}\,\|Z\|_{\mathrm{op}}^3\,. \tag{21}$$

Also,

$$\psi''(0) = \frac{2}{n_r^2}\,\mathrm{tr}(D_p A^2) + \frac{4}{n_r^2}\,\mathrm{tr}(D_p W_r B),$$

which yields

$$\big|\psi''(0)\big| \le \frac{C\,\|D_p\|_{\mathrm{op}}}{n_r^2}\big(\|A\|_{\mathrm{op}}^2 + \|W_r\|_{\mathrm{op}}\,\|B\|_{\mathrm{op}}\big) \le \frac{C}{n_r^2}\big(\|Z\|_{\mathrm{op}}^2 + 1\big). \tag{22}$$

As before, $p^{-1/2}\,\|Z\|_{\mathrm{op}}$ has moments of all orders bounded uniformly in p, and hence

$$\Big(\mathbb{E}\,\|\nabla g_{2,p}\|_2^4\Big)^{1/4} = O(p^{-1/2}), \qquad \Big(\mathbb{E}\,\big\|\nabla^2 g_{2,p}\big\|_{\mathrm{op}}^4\Big)^{1/4} = O(p^{-1}). \tag{23}$$

Combining (20) and (23) with (15), we conclude that, for the full Gaussian argument vector underlying (X, ε, Z),

$$\Big(\mathbb{E}\,\|\nabla L_p\|_2^4\Big)^{1/4} = O(p^{-1/2}), \qquad \Big(\mathbb{E}\,\big\|\nabla^2 L_p\big\|_{\mathrm{op}}^4\Big)^{1/4} = O(p^{-1}). \tag{24}$$

Step 5: asymptotic normality of the linear term

Let

$$s_p^2 := \mathrm{Var}(L_p).$$

By Assumption 6,

$$s_p^2 \asymp p^{-1}. \tag{25}$$

Applying Chatterjee's second-order Poincaré inequality to the smooth Gaussian functional L_p, and using (24)–(25), we obtain

$$d_{\mathrm{TV}}\big(L_p, N(0, s_p^2)\big) \le C\, \frac{\big(\mathbb{E}\,\|\nabla L_p\|_2^4\big)^{1/4}\big(\mathbb{E}\,\big\|\nabla^2 L_p\big\|_{\mathrm{op}}^4\big)^{1/4}}{s_p^2} = O(p^{-1/2}) \to 0.$$

Hence

$$\frac{L_p}{s_p} \Longrightarrow N(0,1). \tag{26}$$

Step 6: completion of the proof

From (11),

$$\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2 = L_p + b_p + o_p(p^{-1/2}),$$

with $b_p = O(p^{-1})$ by (12). Since $s_p \asymp p^{-1/2}$ by (25), it follows that

$$\frac{\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2}{s_p} = \frac{L_p}{s_p} + o_p(1).$$

Combining this with (26) yields

$$\frac{\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2}{s_p} \Longrightarrow N(0,1).$$

In particular,

$$\mathrm{Var}(\widehat{\sigma}_\alpha^2) = s_p^2 + o(p^{-1}), \qquad \mathbb{E}\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2 = O(p^{-1}).$$

Therefore

$$\frac{\widehat{\sigma}_\alpha^2 - \mathbb{E}\widehat{\sigma}_\alpha^2}{\sqrt{\mathrm{Var}(\widehat{\sigma}_\alpha^2)}} \Longrightarrow N(0,1),$$

which proves part (ii).

If in addition $p s_p^2 \to v^2 \in (0,\infty)$, then multiplying the previous display by $\sqrt{p}$ gives

$$\sqrt{p}\,\big(\widehat{\sigma}_\alpha^2 - \sigma_\alpha^2\big) \Longrightarrow N(0, v^2),$$

proving part (iii). This completes the proof. □

Concluding remark

The theorem is a fixed-effect result. The parameter σ_α^2 is deterministic and depends on the realized vector α only through the population marginal signal profile

$$m_j = (\Sigma\alpha)_j^2, \qquad 1 \le j \le p.$$

At no point is α modeled as random. The proof uses two ingredients: first, Wishart moment identities to obtain the exact leading means and the required variance bounds for the reference-panel LD scores; second, Chatterjee's second-order Poincaré inequality to deduce Gaussian fluctuations of the linearized statistic L_p.